Scale selection for monotone functionals via embedded birth–death Markov chains

1 Tridiagonal Inverse

When A is invertible (i.e. all $d_k \neq 0$), the inverse A^{-1} is *tridiagonal*. Its entries can be written explicitly in terms of the differences $d_k = a_k - a_{k-1}$ [1, 7].

Setting $d_k = a_k - a_{k-1}$ (with $a_0 := 0$), the inverse A^{-1} has only three nonzero bands.

Diagonal entries:

$$\left(A^{-1}\right)_{ii} \;=\; \begin{cases} \dfrac{1}{d_i} + \dfrac{1}{d_{i+1}} & \text{for } 1 \le i < N, \\ \dfrac{1}{d_N} & \text{for } i = N. \end{cases}$$

Off-diagonal entries (superdiagonal and subdiagonal):

$$\left(A^{-1}\right)_{i,\,i+1} \;=\; \left(A^{-1}\right)_{i+1,\,i} \;=\; -\frac{1}{d_{i+1}}, \qquad 1 \le i < N.$$

All other entries are zero.

In terms of the original sequence a_k, A^{-1} has the following explicit form:

$$\begin{pmatrix} \dfrac{1}{a_1} + \dfrac{1}{a_2 - a_1} & -\dfrac{1}{a_2 - a_1} & 0 & \cdots & 0 \\ -\dfrac{1}{a_2 - a_1} & \dfrac{1}{a_2 - a_1} + \dfrac{1}{a_3 - a_2} & -\dfrac{1}{a_3 - a_2} & \cdots & 0 \\ 0 & -\dfrac{1}{a_3 - a_2} & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \dfrac{1}{a_{N-1} - a_{N-2}} + \dfrac{1}{a_N - a_{N-1}} & -\dfrac{1}{a_N - a_{N-1}} \\ 0 & \cdots & \cdots & -\dfrac{1}{a_N - a_{N-1}} & \dfrac{1}{a_N - a_{N-1}} \end{pmatrix}.$$

2 Scale Selection for Monotone Functionals f via Embedded Birth–Death Markov Chains

Embedded discrete-time birth–death chain and the critical scale

Setting. Let $a_1 < a_2 < \cdots < a_N$ be a monotonically increasing sequence of real scale parameters, for instance filtration values of a simplicial complex $K(S)$, and let

$$d_i \;:=\; a_i - a_{i-1} \;>\; 0, \qquad i = 2, \ldots, N,$$

denote the consecutive gaps. Assume that a scale similarity matrix $A \in \mathbb{R}^{N\times N}$ built from these scales admits a tridiagonal inverse, and define the discrete Laplacian

$$\mathcal{L} \;=\; -A^{-1}.$$

Then $\mathcal{L}$ has negative diagonal entries, positive off-diagonal entries $1/d_{i+1}$ connecting neighbours $i \leftrightarrow i+1$, and row sums that vanish for interior states. Consequently, $\mathcal{L}$ acts as the generator of a continuous-time birth–death chain on the state space $\{1,\dots,N\}$ with transition rates

$$q_{i\to i+1} \;=\; \frac{1}{d_{i+1}} \;=\; \frac{1}{a_{i+1}-a_i}, \qquad q_{i\to i-1} \;=\; \frac{1}{d_i} \;=\; \frac{1}{a_i-a_{i-1}},$$

so that the birth and death rates are inversely proportional to the gaps between consecutive scales: transitions are fast where the a_k are closely spaced and slow where they are widely spaced. At the boundaries $i=1$ and $i=N$ the chain is subject to *reflection conditions*: state 1 only admits a birth rate $q_{1\to 2} = 1/d_2$, and state N only admits a death rate $q_{N\to N-1} = 1/d_N$.

Embedded discrete-time chain. The embedded (jump) chain of the above continuous-time process is the discrete-time Markov chain on $\{1,\dots,N\}$ with transition probabilities, for interior states $i \in \{2,\dots,N-1\}$,

$$P^{\mathrm{BD}}_{i,i+1} \;=\; \frac{1/d_{i+1}}{1/d_i + 1/d_{i+1}}, \qquad P^{\mathrm{BD}}_{i,i-1} \;=\; \frac{1/d_i}{1/d_i + 1/d_{i+1}},$$

and boundary modifications $P^{\mathrm{BD}}_{1,2} = P^{\mathrm{BD}}_{N,N-1} = 1$ enforcing reflection at the extremes.

Stationary distribution of the embedded chain. Let r_i denote the total outgoing rate from state i in the continuous-time chain, i.e.,

$$r_i \;=\; \begin{cases} \dfrac{1}{d_2}, & i=1,\\[2mm] \dfrac{1}{d_i}+\dfrac{1}{d_{i+1}}, & 2\le i\le N-1,\\[2mm] \dfrac{1}{d_N}, & i=N. \end{cases}$$

The detailed-balance equation for the embedded chain reads

$$\pi^{\mathrm{BD}}_i\, P^{\mathrm{BD}}_{i,i+1} \;=\; \pi^{\mathrm{BD}}_{i+1}\, P^{\mathrm{BD}}_{i+1,i},$$

and, after substitution and telescoping, yields

$$\frac{\pi^{\mathrm{BD}}_{i+1}}{\pi^{\mathrm{BD}}_i} \;=\; \frac{r_{i+1}}{r_i},$$

so that $\pi^{\mathrm{BD}}_i \propto r_i$. The normalised stationary distribution of the embedded birth–death chain is therefore

$$\pi^{\mathrm{BD}}_i \;=\; \frac{r_i}{\sum_{j=1}^{N} r_j} \;=\; \frac{\frac{1}{d_i}+\frac{1}{d_{i+1}}}{\sum_{j=1}^{N}\left(\frac{1}{d_j}+\frac{1}{d_{j+1}}\right)}, \qquad 2\le i\le N-1, \tag{1}$$

with the boundary modifications $\pi^{\mathrm{BD}}_1 \propto 1/d_2$ and $\pi^{\mathrm{BD}}_N \propto 1/d_N$ indicated by the convention $r_1 = 1/d_2$, $r_N = 1/d_N$.

Identification of the critical scale. Since $\pi_i^{\mathrm{BD}} \propto r_i = 1/d_i + 1/d_{i+1}$, the stationary distribution is determined entirely by the local increments of the complexity functional around state i. It is *small* exactly at those indices i where both d_i and d_{i+1} are *large*, i.e., where the functional f undergoes sharp jumps on *both* sides of the scale K_i; it is *large* at indices where f varies slowly, i.e., where the filtration crosses a plateau of nearly constant complexity.

Locally large jumps of f signal a *topological transition* in the filtration: the complex K_{i-1} and K_i are structurally distinguishable under the chosen complexity functional, so the scale a_i (equivalently, the filtration step between K_{i-1} and K_{i+1}) marks a regime boundary between topologically distinct phases. The critical scale is therefore naturally identified as

$$i^\star \;=\; \operatorname*{arg\,min}_{i \in \{1,\dots,N\}} \pi_i^{\mathrm{BD}}. \tag{2}$$

This criterion has a direct interpretation at the level of the complexity functional, independent of any spectral analogy: (2) selects the filtration step at which the pair of increments $d_{i^\star}, d_{i^\star+1}$ is simultaneously largest, i.e., the step across which the functional f exhibits its most pronounced local variation on both sides. Such a step separates two regions of the filtration on which f varies slowly — plateaux of approximately constant topological complexity — and therefore corresponds to the most informative scale at which to analyse the complex. The recipe is a topologically principled analogue of the classical *elbow rule* for scree plots in PCA, with the role of eigenvalue gaps played by jumps of the complexity functional along the filtration.

Remark on decreasing sequences. If the input sequence $(a_j)_{j=1}^N$ is monotonically *decreasing* rather than increasing, all of the above applies verbatim after replacing the differences d_j by their absolute values; equivalently, one multiplies each d_j by -1, which leaves the rates $1/|d_j|$, the embedded transition probabilities, the stationary distribution (1), and the critical-scale identification (2) unchanged. Hence the methodology is invariant under reversal of the enumeration of scales, and no separate treatment of the decreasing case is required.

2.1 Continuous-time analogue: time-changed Brownian motion and the critical scale

Setting. Let $a\colon [0,1] \to \mathbb{R}$ be an absolutely continuous, non-decreasing function on the filtration parameter interval $[0,1]$, with derivative $a'(s) \geq 0$ for almost every s. The function a plays the continuous analogue of the discrete sequence $(a_j)_{j=1}^N$ of complexity-functional values along the filtration, and $a'(s)$ is the continuous analogue of the gaps d_j: where a' is large, the complexity functional varies rapidly with respect to the filtration parameter, corresponding to large discrete gaps; where a' is small, the functional is nearly constant, corresponding to plateaux in the discrete picture.

Let $(W(t))_{t\geq 0}$ be a standard Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$, and define the Itô integral process

$$X(t) \;=\; \int_0^t \sqrt{a'(s)}\, dW(s), \qquad 0 \leq t \leq 1. \tag{3}$$

Equation (3) is a standard Itô stochastic integral against Brownian motion [8, 5], well defined since $a' \in L^1([0,1])$ implies $\sqrt{a'} \in L^2([0,1])$.

Quadratic variation and Itô representation. The process X is a continuous $(\mathcal{F}_t)$-martingale starting at $X(0) = 0$. By Itô's isometry,

$$\mathbb{E}[X(t)^2] \;=\; \int_0^t a'(s)\,ds \;=\; a(t) - a(0),$$

and, more generally, the quadratic variation of X is

$$\langle X\rangle_t \;=\; \int_0^t a'(s)\,ds \;=\; a(t) - a(0), \qquad 0 \le t \le 1. \tag{4}$$

This identifies X as a continuous martingale whose intrinsic clock is the complexity-functional-value $a(t)$.

Dambis–Dubins–Schwarz representation. The Dambis–Dubins–Schwarz theorem [3, 4] (see also [8, Theorem V.1.6] or [5, Theorem 3.4.6]) states that every continuous $(\mathcal{F}_t)$-martingale M with $M(0) = 0$ and $\langle M\rangle_\infty = \infty$ can be represented as a time-changed Brownian motion with respect to its own quadratic variation clock. Applying this to X from (3) yields the existence of a standard Brownian motion $\tilde{B}$ (possibly on an enlargement of the original probability space) such that

$$X(t) \;=\; \tilde{B}\big(a(t) - a(0)\big), \qquad 0 \le t \le 1. \tag{5}$$

Equivalently, writing $u = a(t) - a(0)$ for the new clock variable, the process X runs according to $\tilde{B}$ on the interval $[0, a(1) - a(0)]$ of the u-clock. The time change $t \mapsto a(t)$ is the unique continuous, non-decreasing map aligning the filtration clock t with the quadratic-variation clock of X.

Interpretation of the time change. The representation (5) makes explicit the sense in which the complexity functional a drives the "speed" of X. Where $a'(s)$ is large (the filtration passes quickly through a topologically active regime), a small increment in s produces a large increment in the u-clock and hence X moves rapidly, tracing out a long path of $\tilde{B}$. Where $a'(s)$ is small (the filtration rests on a plateau of the complexity functional), the u-clock barely advances, and X is nearly constant over such intervals. The process X is therefore a Brownian motion in the intrinsic clock of the filtration's complexity, and the physical filtration parameter s supplies the external clock against which this intrinsic motion is played.

Occupation times and local time. Let $L^X(t, \ell)$ denote the local time of X at level ℓ up to time t, in the sense of [8, Chapter VI] or [5, Chapter 3.7]. By the occupation-times formula, for any Borel function $g\colon \mathbb{R} \to [0, \infty)$,

$$\int_0^t g(X(s))\,d\langle X\rangle_s \;=\; \int_{\mathbb{R}} g(\ell)\,L^X(t,\ell)\,d\ell. \tag{6}$$

Combining (4) with (6) yields

$$\int_0^t g(X(s))\,a'(s)\,ds \;=\; \int_{\mathbb{R}} g(\ell)\,L^X(t,\ell)\,d\ell, \tag{7}$$

which identifies $L^X(t, \cdot)$ as the *Brownian* local time of $\tilde{B}$ evaluated at the time-changed clock:

$$L^X(t,\ell) \;=\; L^{\tilde{B}}\big(a(t) - a(0),\, \ell\big). \tag{8}$$

The local-time profile of X at a fixed horizon $t = 1$ is therefore a Brownian local-time profile on the functional-value interval $[a(0), a(1)]$.

Clock-time occupation density. To connect with the scale-selection criterion, it is the *clock-time occupation* of X at a level ℓ — the real (filtration) time that X spends near ℓ — that matters. By the change of variable $u = a(s)$, $du = a'(s)\,ds$, and hence $ds = du/a'(a^{-1}(u))$ on regions where a' is strictly positive, we obtain

$$\int_0^1 \mathbf{1}_{\{X(s)\approx\ell\}}\,ds \;\approx\; \frac{L^{\tilde{B}}(a(1)-a(0),\,\ell)}{a'\big(a^{-1}(\ell)\big)}, \tag{9}$$

valid for any level $\ell \in \big(a(0), a(1)\big)$ in the range of a. The clock-time occupation density at level ℓ is therefore the Brownian local time of $\tilde{B}$ at ℓ *divided by* the derivative a' evaluated at the preimage $s = a^{-1}(\ell)$. The Brownian local-time factor in (9) is universal (it depends only on $\tilde{B}$, not on a), so the a-dependent modulation is entirely captured by the factor $1/a'(a^{-1}(\ell))$.

Continuous-time analogue of the scale-selection rule. In the discrete setting, the embedded birth–death chain has stationary distribution

$$\pi_i^{\mathrm{BD}} \;\propto\; \frac{1}{d_i} + \frac{1}{d_{i+1}} \;\approx\; \frac{2}{d_i} \quad \text{for smoothly varying gaps,}$$

and the critical-scale criterion is $i^\star = \arg\min_i \pi_i^{\mathrm{BD}}$. In the continuous limit, with $d_i \approx a'(s_i)\,\Delta s$ for a local refinement Δs, one has $\pi_i^{\mathrm{BD}} \propto 1/a'(s)$, and the discrete argmin passes to the continuous

$$s^\star \;=\; \arg\max_{s\in[0,1]} a'(s), \tag{10}$$

the filtration time at which the complexity functional varies most rapidly. Equivalently, in terms of the associated level $\ell^\star = a(s^\star)$ and the clock-time occupation (9),

$$\ell^\star \;=\; \arg\min_{\ell\in(a(0),a(1))} \frac{L^{\tilde{B}}(a(1)-a(0),\,\ell)}{a'\big(a^{-1}(\ell)\big)} \Big/ L^{\tilde{B}}(a(1)-a(0),\,\ell). \tag{11}$$

That is, $\ell^\star$ is the level at which the clock-time occupation density of X, normalised by the (universal) Brownian local time of $\tilde{B}$, is smallest.

Physical interpretation. The criterion (10)–(11) admits a direct interpretation. Large $a'(s)$ means that the complexity functional changes rapidly with the filtration parameter, so the time-changed Brownian motion X transits the corresponding level $\ell = a(s)$ quickly, spending little clock-time there relative to how much Brownian local time it accumulates in the intrinsic clock. Conversely, small $a'(s)$ (a plateau of the functional) means that X accumulates a great deal of clock-time at the corresponding level, but this is merely the filtration resting on a region of topological stasis.

The critical scale is therefore *not* the level of longest local time in clock time — such levels are plateaux, where the complex is topologically stationary under the chosen functional — but rather the level at which the clock-time occupation, corrected for the underlying Brownian local time, is minimised. This continuous-time reformulation clarifies the topological content of the discrete argmin recipe: it selects the filtration instant of *steepest ascent* of the complexity functional, at which the complex undergoes its most significant local change. The apparent paradox — argmin of a stationary distribution identifying a critical scale — is resolved once one observes that the discrete chain is constructed from the inverse A^{-1} of the scale similarity matrix, so that states with low π_i^{BD} are precisely those connected to their neighbours by slow rates, i.e., those at which the clock-time of X moves fastest through the functional landscape.

Decreasing complexity functionals. If the complexity functional is chosen such that a is monotonically *decreasing*, the construction (3) requires a' to be replaced by $|a'|$ in the integrand, equivalently by the replacement $a \mapsto -a$ that leaves all quadratic variations, local times, and occupation-time formulae invariant. The criteria (10) and (11) then select $\arg\max_s |a'(s)|$, the filtration time of largest absolute rate of change of the functional, which is the intended topologically-critical scale. Hence the methodology is invariant under the monotonicity convention for the complexity functional.

References. The Itô integral construction (3) and its quadratic variation (4) are classical and can be found in any standard reference on stochastic calculus; we follow the notation of [8] and [5]. The Dambis–Dubins–Schwarz representation (5) is the central result of [3, 4]; textbook treatments are in [8, Theorem V.1.6] and [5, Theorem 3.4.6]. The theory of Brownian local time used in (6)–(8) goes back to [6] and [9], with the modern semimartingale framework due to Tanaka and Meyer; see [8, Chapter VI] for a self-contained account, and [2] for an encyclopedic handbook of formulae on Brownian local times and related occupation-time quantities.

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